

Jonathan (Jack) Simpson
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ME

I love problem solving, digital tech, all things science and history, flying stuff, watersports, caffeine, and obnoxiously colored socks. Always striving to learn the newest tools; diversifying and advancing my skills as much as humanly possible. I strive to embrace the discomfort of ignorance as a road to the quickest learning. Appreciating the irony of building the tools that will replace us. I will happily talk in great depth about my projects - past, present and future.

LANGUAGES

Javascript, Typescript, HTML, CSS, Python, GLSL, Human English, B1 German

LIBRARIES / APIs

WebGL, React, Three, R3F, D3, Node, Vue

SOFTWARE

Figma, VS Code, Maya, Realflow, Nuke, Vray, Adobe Suite, Git, Slack, Jira, OSX, Windows, Linux

CLIENTS

NASA, Disney, MagMutual, Intuit, Society for Neuroscience, Macy's, HP, Amazon, Hyundai, Kmart, Samsung, NFL, Extell, VH1, ServPro, Carmax, Oxygen, WCS, Saatchi&Saatchi, Kellogg's, Budweiser, Leo Burnett, DDB, Euro RSCG

WORK HISTORY

Feb 2021 – Sep 2024 **Senior Web Visualization Engineer**
NASA Jet Propulsion Lab

- Developing NASA's "Eyes" suite of educational web applications to expand public knowledge
- Visualizing our solar system using a range of spacecraft, planet and asteroid tracking data
- Designing intuitive UX wireframes and Figma prototypes for optimized user learning
- Utilizing React alongside proprietary frameworks to integrate UI functionality and immersive 3D navigation
- Directing a dynamic team of developers, overseeing design conception, agile development and project delivery
- Pioneering research in 3D web animation techniques, setting new standards for online visual engagement
- Recognized with Webby and NASA Honor Award in 2023 for "Eyes on Asteroids"

Apr 2018 – Nov 2020 **Senior Web Developer**
MJD Interactive / Valtech

- Building React app WebGL features for a variety of US clients
- Integrating Vue frontend UIs with complex backend theme park architecture
- Designing custom interactive particle systems using GLSL shaders
- Using D3 and extending three.js library
- Generating custom geometry with interactive hyper-real materials
- Working full-time remotely
- 2 Webby Awards

Jan 2017 – Jun 2019 **Interactive WebGL Developer**
Society for Neuroscience

- Building an interactive 3D brain module for a public neuroscience education website
- Developing techniques to smoothly subdivide 3D polygon buffer arrays using web workers to reduce load times
- Animating between light map textures in GLSL shaders
- Working closely with a medical illustrator and re-modeling public brain scan data into accurate organic polygonal models

May 2016 – Nov 2017 **Lead Interactive WebGL Developer**
Macy's

- Lead 3D developer for an interactive web application for the Thanksgiving Day parade
- Building custom python scripts for geometry exports
- Creating custom JavaScript navigation and animation solutions
- Developing advanced animation and shader concepts with GLSL
- Applying performance optimization techniques and graceful fallback options
- Directing a team of 3D artists and collaborating with lead front-end developers
- 3 Webby Awards

Dec 2015 – Feb 2017 **Lead Interactive WebGL Developer**
The Molecule

- Lead 3D developer for a large interactive experiential real estate sales project
- Modifying, structuring and managing 3D assets
- Creating WebGL compatibility for 360 degree projections
- Building custom scripts with parent structures and optimized instancing
- Constructing JavaScript functions and modules to manipulate 3D geometry
- Developing GLSL shaders and animation techniques

Sep 2015 – Nov 2015 **WebGL Developer**
Huge Inc & HyperHyper

- Creating custom python export scripts for geometry, bones, and animation
- Utilizing three.js library to test and develop interactive animation characters
- Constructing JavaScript functions and modules to manipulate 3D characters
- Directing and team of 3D artists and game developers
- Working with GLSL shaders
- Optimizing animation performance for functionality with mobile browsers

Apr 2015 – Oct 2015 **Lead 3D Artist**
Advanced Thermal Therapeutics

- Building and animating 3D hyper-real medical product visualizations
- Cloth and particle simulation
- Outsourcing render farm management

July 2015 – Aug 2015 **3D Generalist**
Blacksmith VFX

- Modeling, texturing, animating, lighting, and rendering CG elements
- Compositing photo-real 3D with live action footage

Jan 2012 – Jun 2015 **3D Technical Director**
Eyeball NYC, New York

- Technical consulting for clients and creative directors
- Programming and structuring network pipelines
- Coding custom automation scripts for Maya and Nuke
- Shot design and layout
- 3D modeling, texturing, lighting, animation and rendering
- Dynamic simulation of cloth, water, explosions, code-driven particles
- Compositing 2D and 3D elements
- Lead front-end web development using Backbone and jQuery frameworks
- Drone operating for aerial videography 3D photogrammetry mapping

Oct 2011 – Dec 2011 **3D Artist**
Eyeball NYC, New York

- 3D modeling, texturing, animation, simulation, lighting and rendering for post-production 3D elements
- Assisting lead compositors with paint and roto-scoping, tracking, animation and slap composition
- Research and development into 3D kinematics, dynamics, fluid simulation, vehicle rigging solutions, and rendering techniques
- Advancing code skills to make production pipeline tools

Jul 2011 – Sep 2011 **Technical Director Intern**
Framestore, London

- Creating photorealistic materials and shading networks, computer-generated studio lighting and rendering high-resolution images
- Re-structuring staff training pipeline for Digital Imaging subdivision

Sep 2009 – Aug 2010 **Animation Director**
Animated Storyboards, London and New York

- Directing animated test commercials (2D and 3D animation styles) for American and European clients
- Consulting with global marketing agencies
- Video editing, color correction, exporting, and compressing for delivery
- Shot design and layout
- Directing photography and 3D motion capture shoots
- Working on up to 8 simultaneous projects
- Compiling word document briefs with reference videos for 3D artists
- Developing company workflow and pipeline
- Acting in a motion-capture suit

EDUCATION

Sep 2010 – Aug 2011	Kingston University, London, UK MSc 3D Computer Generated Imagery
Sep 2006 – Jun 2009	Leeds Metropolitan University, Leeds, UK BSc Animation Technology and Special Effects
Sep 2004 – Jun 2006	Truro College, Cornwall, UK A-Level Mathematics (Grade A) A-Level Physics (Grade A) National Diploma in Media (Double Merit)
Sep 1999 – Jun 2004	Penair, Cornwall, UK 11 GCSEs (8 A*s, 1A, 2Bs)

REFERENCES

Jason Craig
Producer at NASA JPL
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John Frenette
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